Week 3 Lecture Note

Topic: Training the CNN and Checking Performance in Google Colab

1. Objectives

(1) Understand what it means to train a model; Learn the meaning of **epoch, loss, and accuracy**.

(2) Run **model.fit()** to start actual learning; Observe how the model improves over time,

2. Key Concepts

(1) What is an Epoch?

An epoch means one full round of learning using the whole training dataset once. An epoch is like one full study session over all practice questions.

Epoch 1 = the model sees all training images one time

Epoch 2 = the model sees them all again

Epoch 3 = one more full round

(2) What is Loss?

Loss shows how wrong the model is: **high loss** = the model is making larger mistakes; **low loss** = the model is making smaller mistakes

Loss is not the same as accuracy; the model tries to minimize loss during training

(3) What is Accuracy?

Accuracy shows how often the model predicts correctly; accuracy = 0.85 means 85% correct

Loss = how wrong. **Accuracy** = how often correct

(4) Saving the Model

Saving the model means **storing what the AI has learned** so you can use it later without training again. In your class, after `model.fit()`, the CNN has adjusted many internal numbers called **weights**. So when you run: `model.save("pneumonia_cnn_week3.h5")`

It includes the **model structure,** the **learned weights, and** often the **training configuration**.

If you save it, you do **not** need to retrain from the beginning